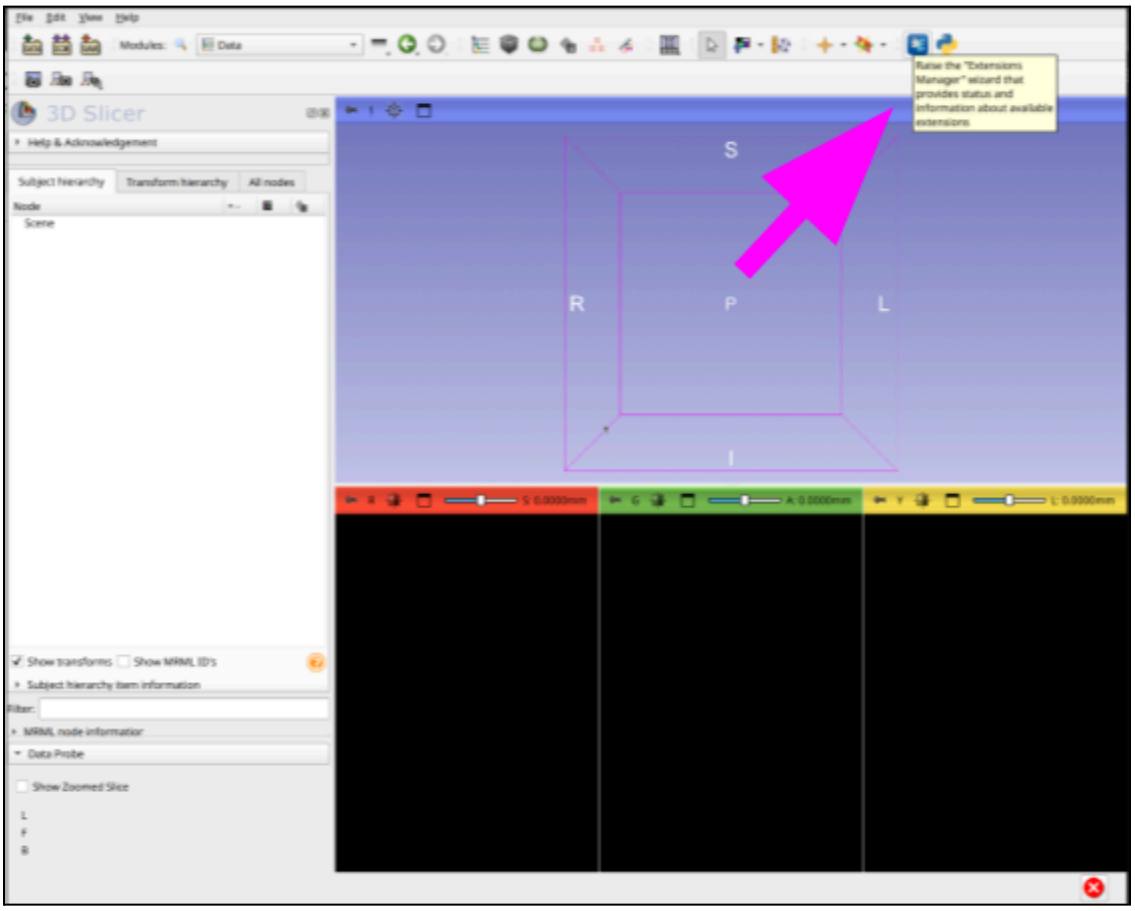
FLMNH Digital Imaging Workshop - 3D GMM in 3D Slicer - Katherine Wolcott
Tuesday, October 27th, from 2:00 – 4:00 pm in Dickinson Hall

Note: **Highlighted text** contains steps where you will interpret morphology. Pay extra attention to these when working with your own data.

Steps

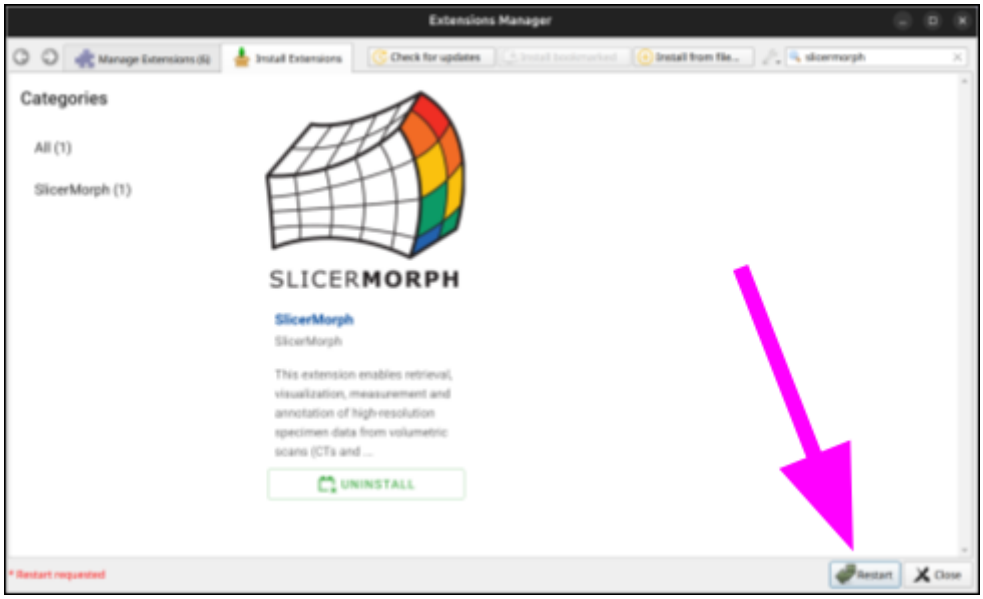
1. Install 3D Slicer
    a. https://download.slicer.org/
2. Open 3D Slicer
3. Open the Extensions Manager module (puzzle piece icon in main navigation bar)



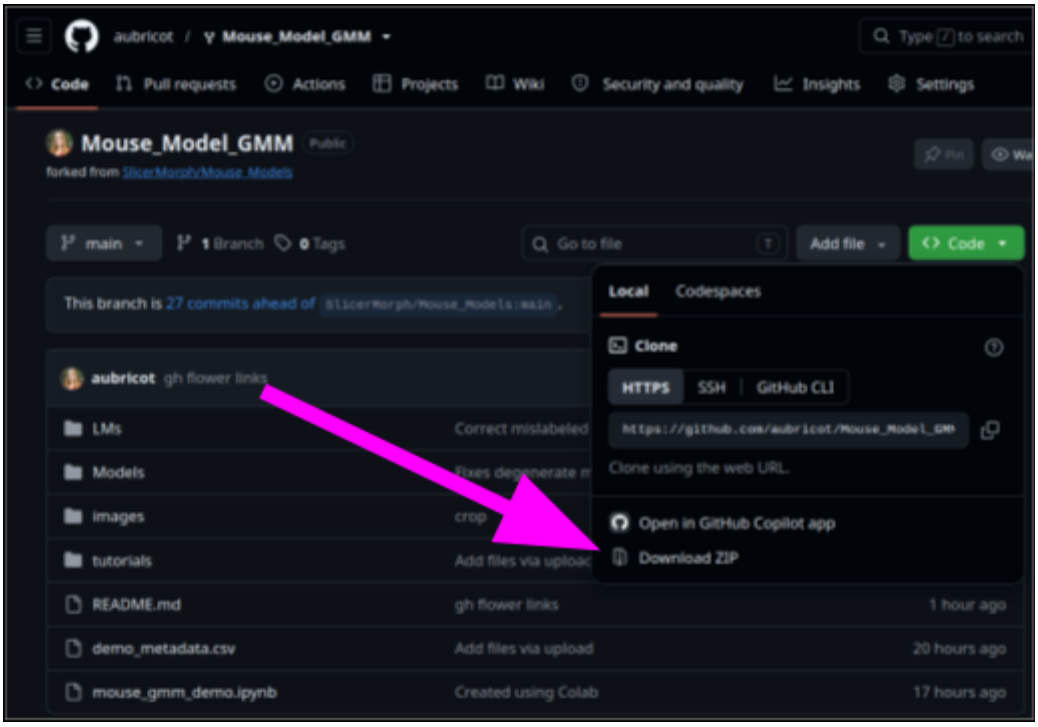
4. Install SlicerMorph
    a. Click “Install Extensions” tab
    b. Search for “SlicerMorph”
    c. Click “Install”



    d. Click “Restart” to restart Slicer and begin using SlicerMorph



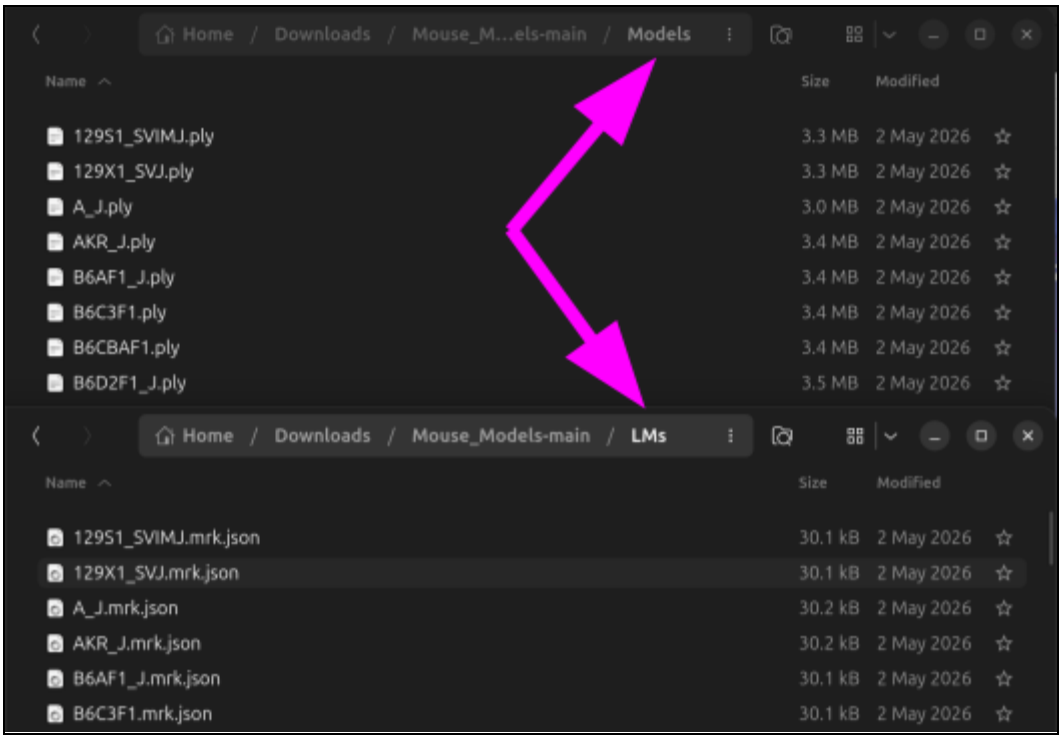
    e. Detailed instructions
        i. https://github.com/SlicerMorph/SlicerMorph#installation
5. Leave 3D Slicer
6. Open your web browser
7. **Download sample data from the tutorial GitHub repo**
    a. Navigate to the Mouse_Models tutorial repository
        i. https://github.com/aubricot/Mouse_Model_GMM/
    b. Click the “Code” dropdown menu
    c. Click “Download Zip”



8. Extract the files for GMM
    a. Locate “Mouse_Models-main.zip” in your Downloads folder
    b. Right click and extract the file (varies by operating system)
    c. Open Mouse_models-main folder



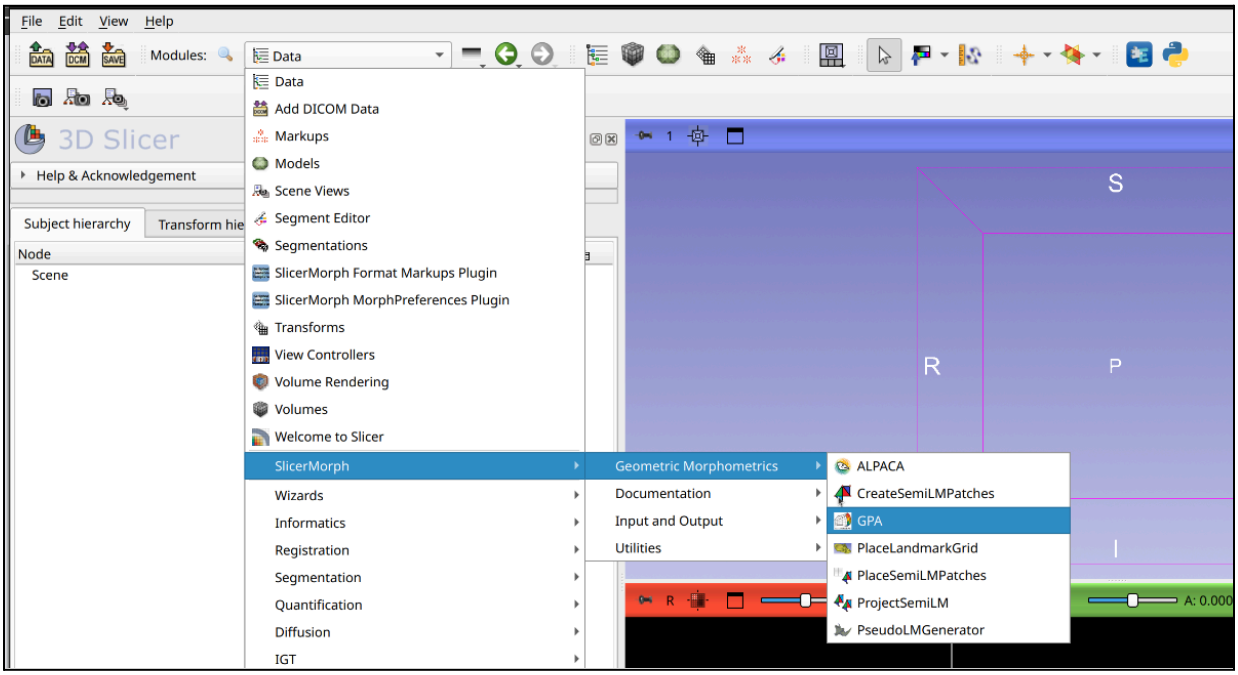
9. **Inspect the mouse data structure - Most 3D geometric morphometrics pipelines require your data to follow this format**
    a. Directory structure
        i. Main directory (Mouse_models-main)
            1. LMs (all landmarks in one folder)
            2. Models (all models in one folder)
    b. File names
        i. Landmark and model files should only differ in their extension
            1. Example
                a. LMs/A_J.mrk.json
                b. Models/A_J.ply



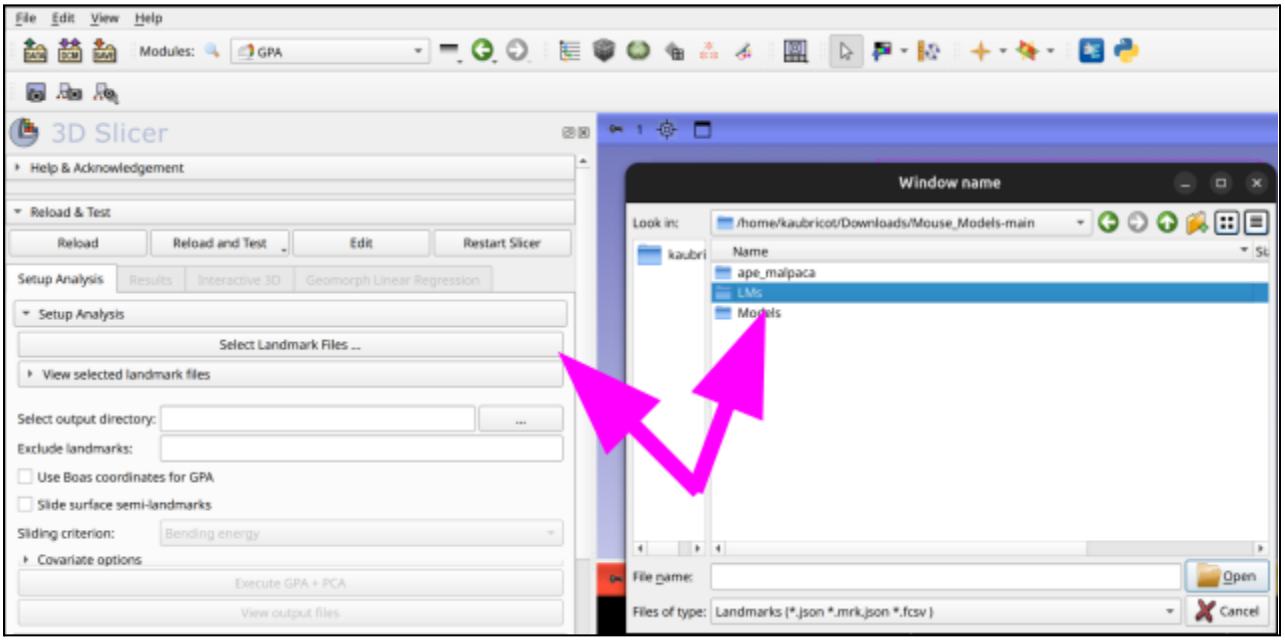
c. When creating your own GMM data, be sure to keep this structure

10. Return to 3D Slicer

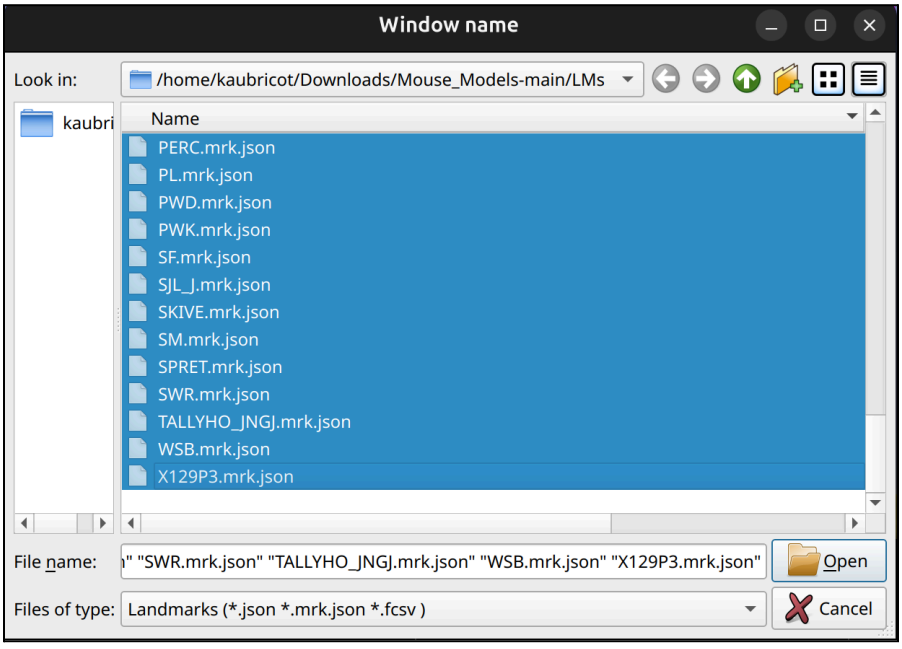
11. Open the GPA submodule from SlicerMorph
    a. Go to the Module List dropdown
    b. Go to SlicerMorph
        i. Geometric Morphometrics
            1. GPA



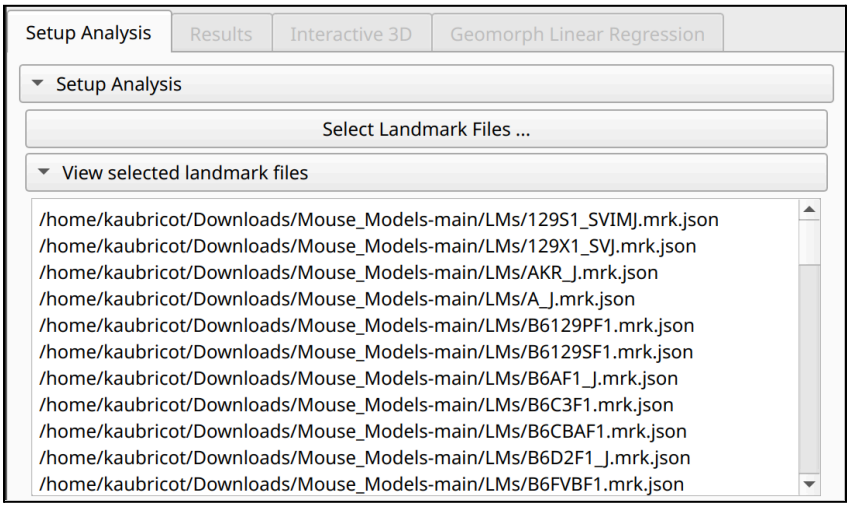
12. Load the landmark files
    a. Click “Select Landmark Files”
    b. Choose the “LMs” folder you downloaded from GitHub



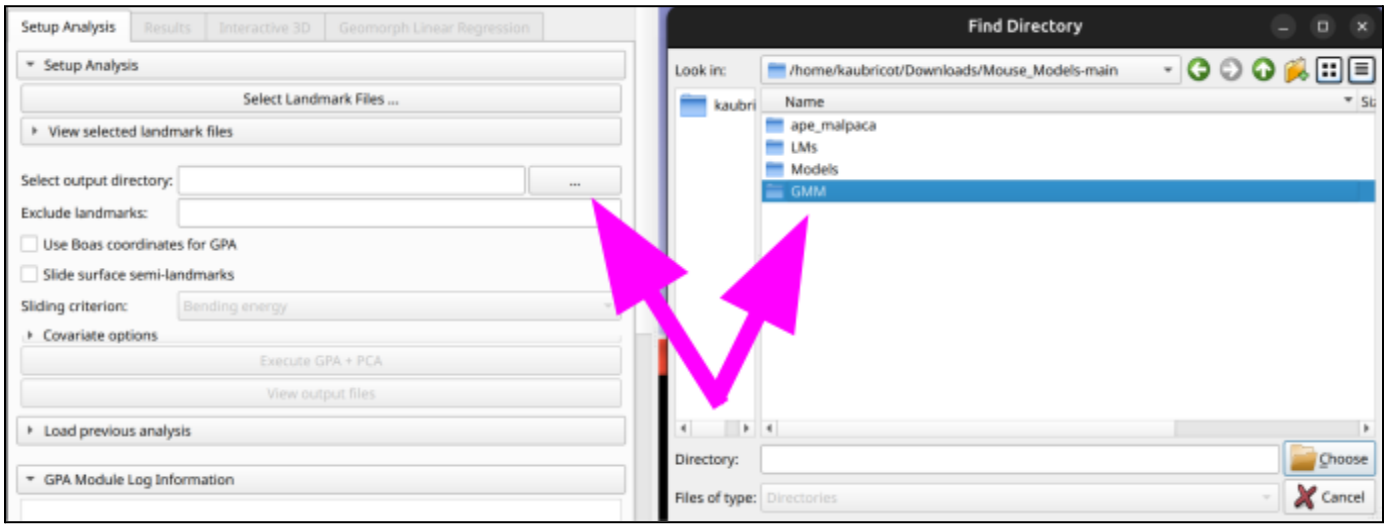
    c. Select all files in the “LMs” folder
    d. Click Open



13. Click the “View selected landmark files” tab to check that they loaded

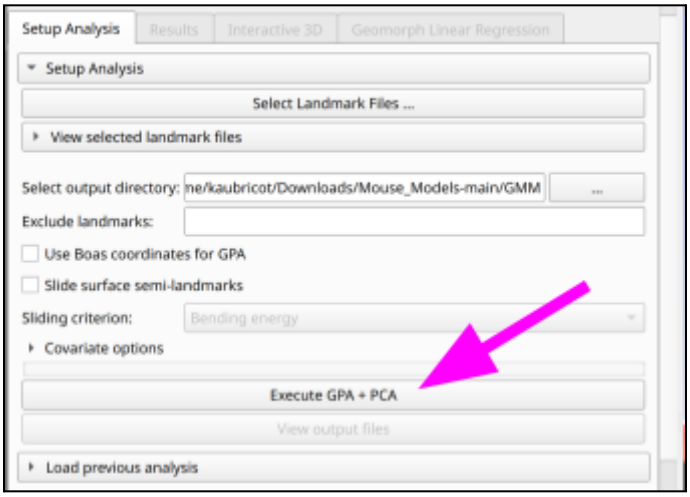


14. Click “Select output directory”
    a. Navigate to “Mouse_Models-main”
    b. Create a new folder, “GMM”
    c. Select “GMM” and hit “Choose”

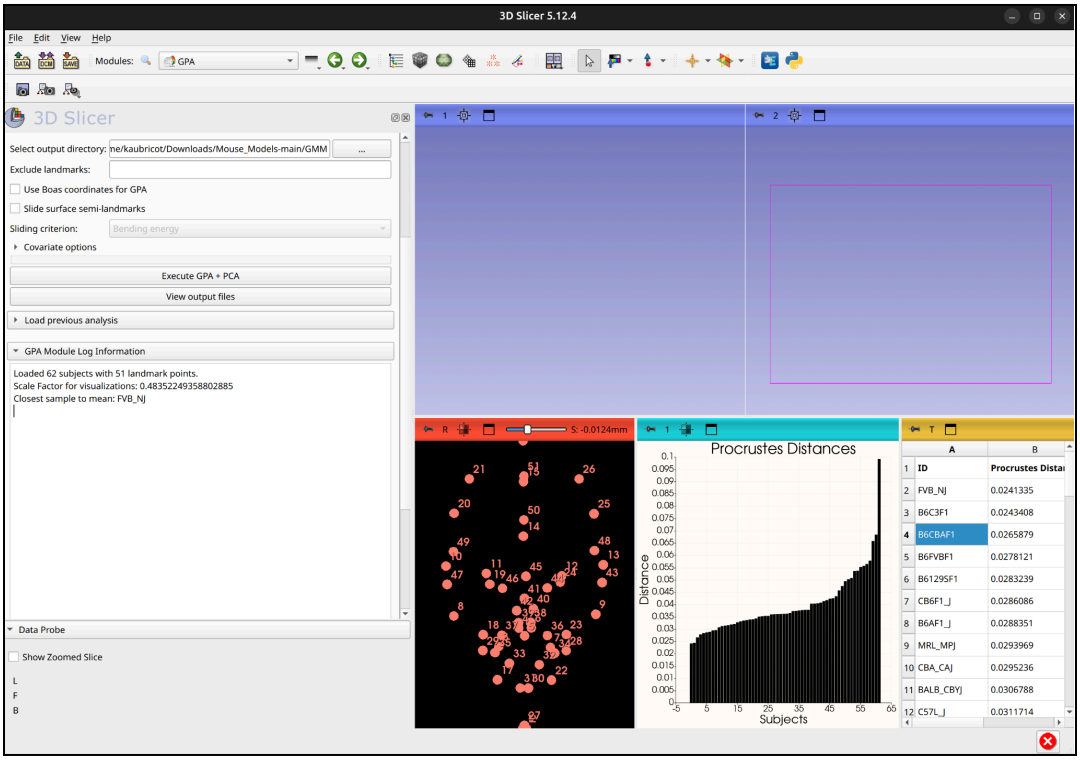


15. Click “Covariate options”
    a. Select covariates table: demo_metadata.csv
        i. This is a table with mock metadata for this tutorial
        ii. This is used to color PCA points by species, age, collection locality, etc.

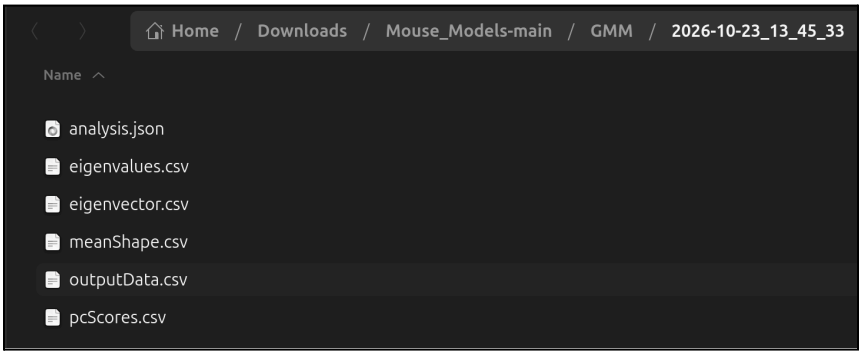
16. Click “Execute GPA + PCA”
    a. GPA is Generalized Procrustes Analysis
        i. It is a superimposition strategy to align and scale meshes
        ii. This minimizes position and size variation, so GMM analysis can focus on shape differences
    b. PCA is Principal Component Analysis
        i. It is a covariance analysis to condense landmarks into independent shape axes, which form the morphospace used to plot and compare specimens



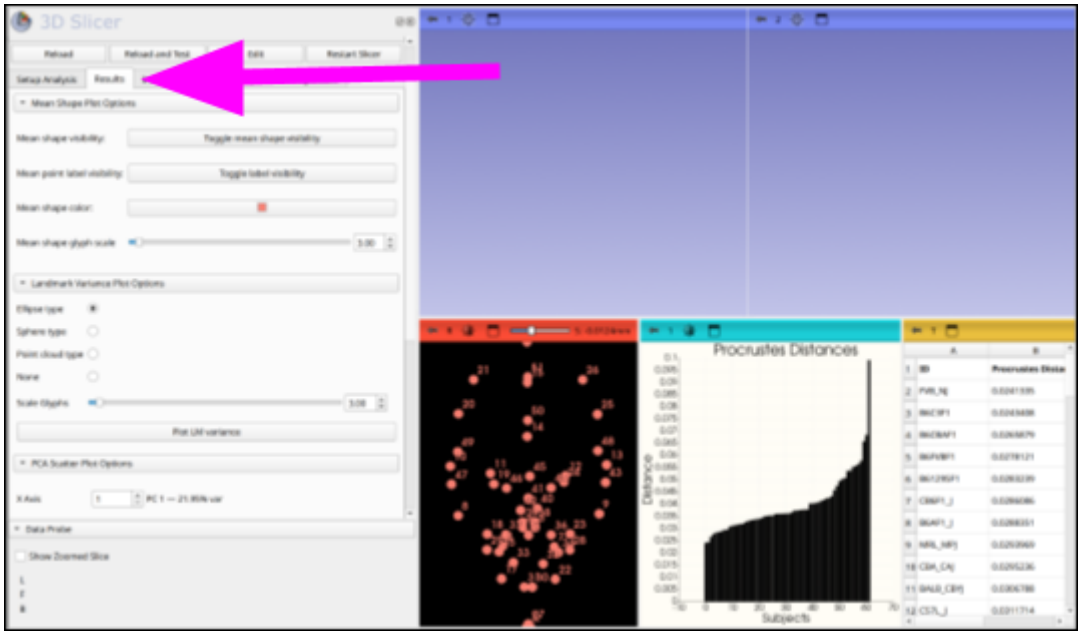
17. Confirm that GPA + PCA succeeded
    a. You should see outputs in the “GPA Module Log Information” tab, and in the three viewer windows on the bottom right side of the screen



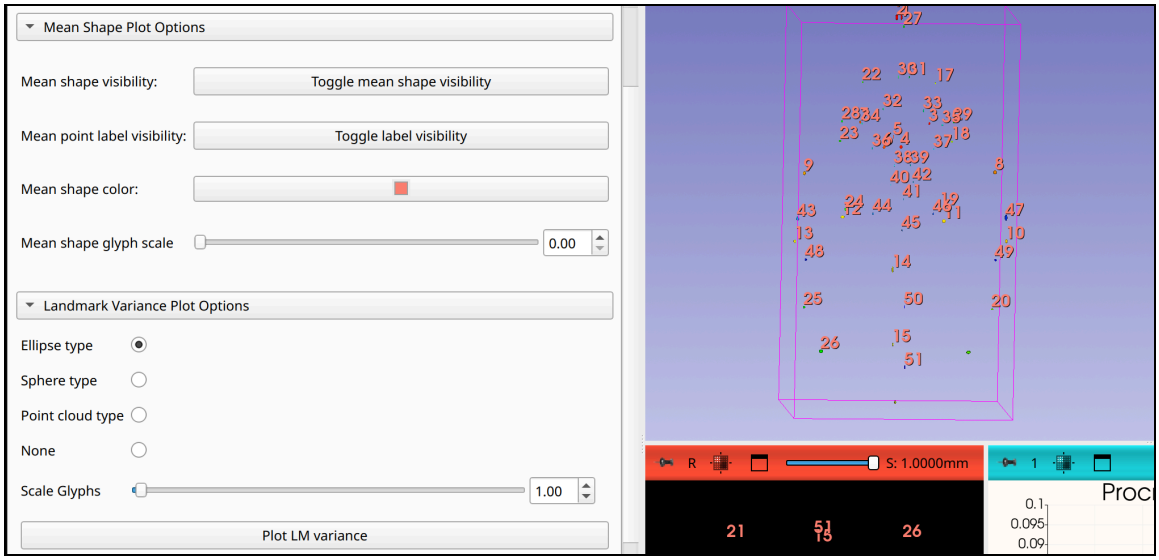
b. To see output files, click “View Output Files”
    i. These files can be loaded into Slicer in a new session to resume analysis

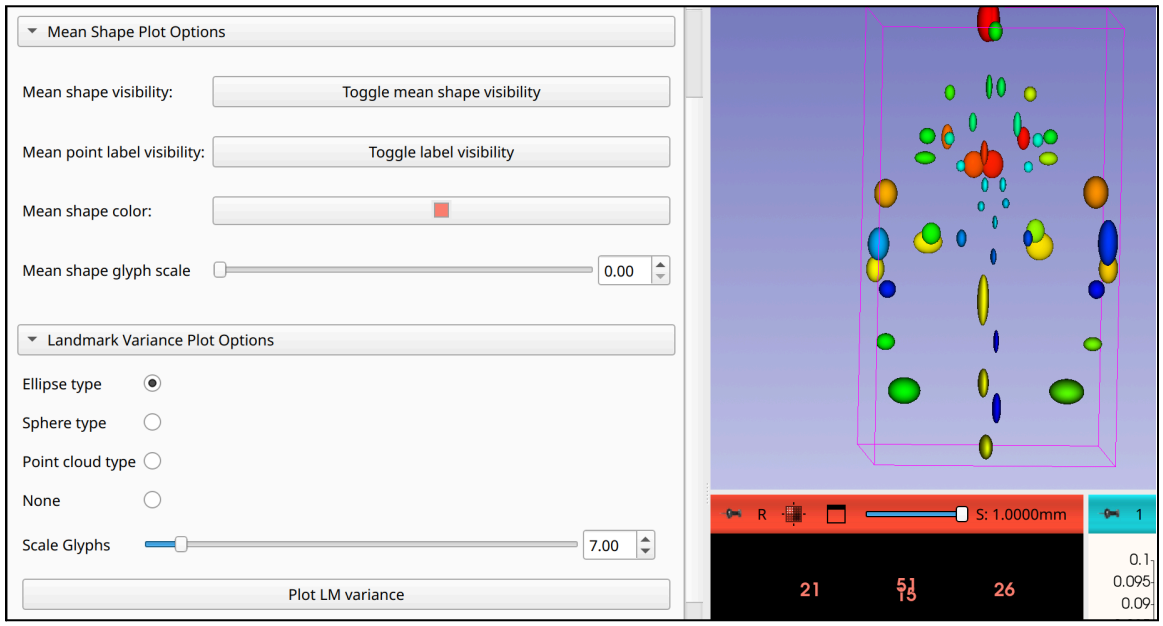


18. Switch to the “Results” tab

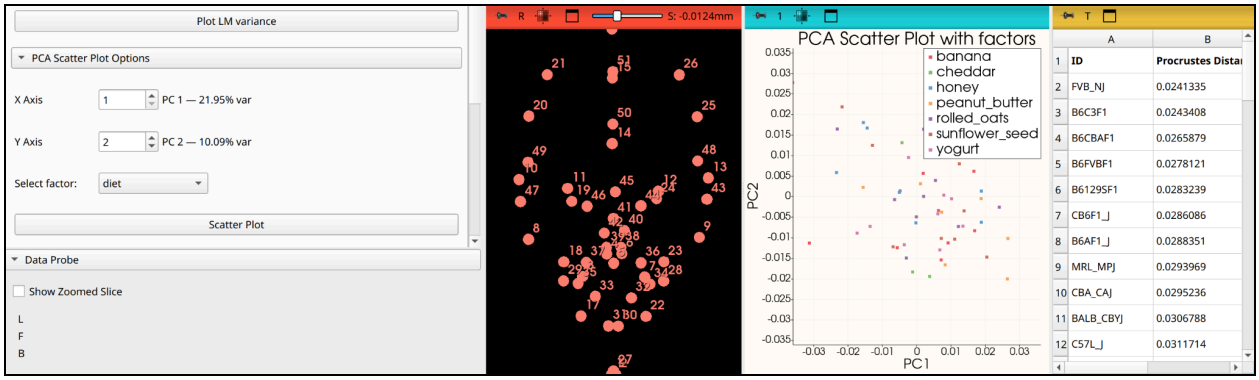


19. Plot landmark variance
    a. This plot shows how much variation is present in your dataset at each landmark
    b. Click “Plot LM variance”
        i. You may need to use the “Center view” button and zoom in/out (scroll wheel) until the landmark variance points show up in the 3D viewer
    c. Inspect which landmarks are most and least variable in XYZ directions
        i. If you don’t see landmark labels on the 3D plot, adjust settings below
            1. Click and rotate the landmark variance points
            2. Adjust “Scale Glyphs” to show labels on points
        ii. For example, landmark 14 (yellow oval along skull midline) is long and skinny
            1. It varies anteroposteriorly, but not much laterally
        iii. Identify another landmark variance plot and try to morphologically interpret what its shape means





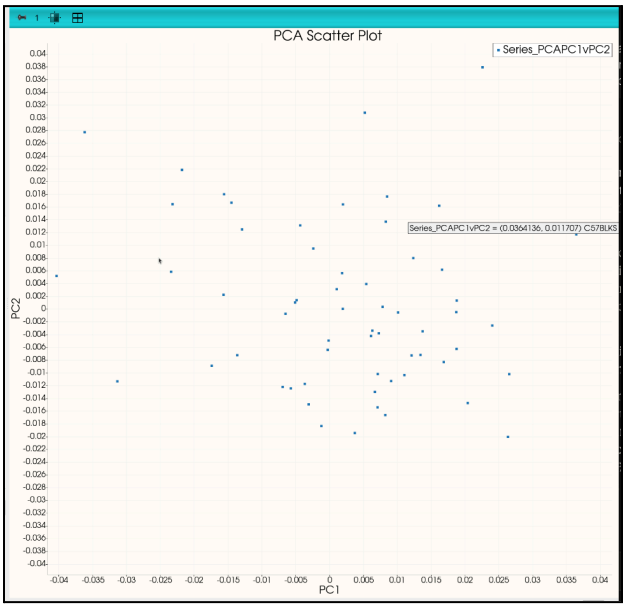
20. Plot the PCA scatterplot
    a. Plot PC1 v PC2
        i. X Axis: PC 1
        ii. Y Axis: PC 2
        iii. Select factor: diet
            1. You can color points by diet, species, age, collection locality, etc.
        iv. Click “Scatter Plot”
    b. Inspect explained proportion of variance values
        i. PC1: 21.95%
        ii. PC2: 10.09%

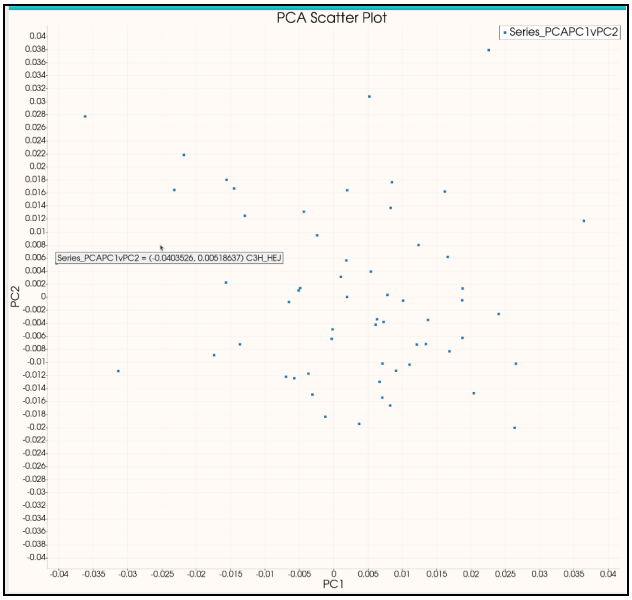


c. Inspect PCA plot in blue window on right
   i. Click “Maximize view” icon to make the PCA plot full screen
   ii. Zoom in and out with your scroll wheel

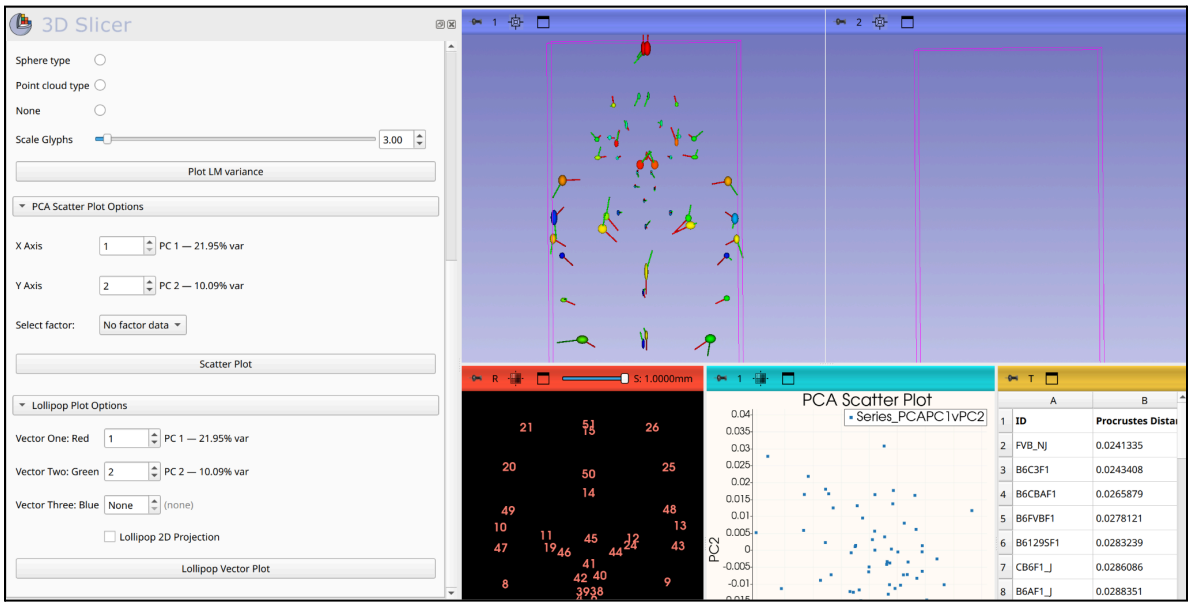
d. Inspect which points are close and far in PC morphospace

   i. Points that are closer together are more similar in shape
   ii. Separation along a PC means the specimens differ in the trait for that axis
      1. For example, specimen C57BLKs is the max shape for PC1 (right side of plot) and specimen C3H_HEJ is the min shape (left side)
      2. These are the min and max of the trait captured by PC1
         a. If PC1 captures elongation, the min shape would be short and the max shape would be long
   iii. Find the min and max shapes of PC2
      1. What shape trait might PC2 capture? Inspect the min and max shapes and make a guess
   iv. Click “Restore View Layout” icon to return to multipanel view

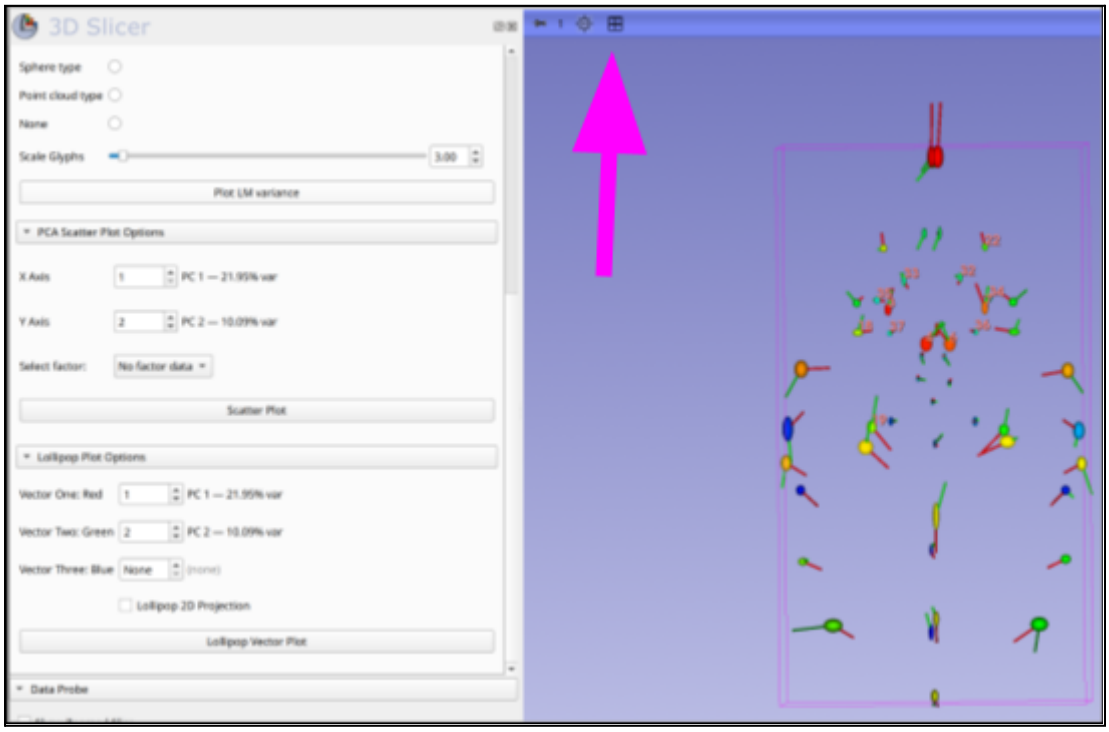




21. Plot lollipop vectors
    a. Lollipop vectors show shape change along trajectories
        i. In 3D Slicer, they are shown by PC, allowing you to show variance for a landmark along PC's
    b. Look at lollipop vectors for PC1 v PC2
        i. Vector One: PC1
        ii. Vector Two: PC2
        iii. Click "Lollipop Vector Plot"



c. Click “Maximize view” icon to make lollipop plot full screen



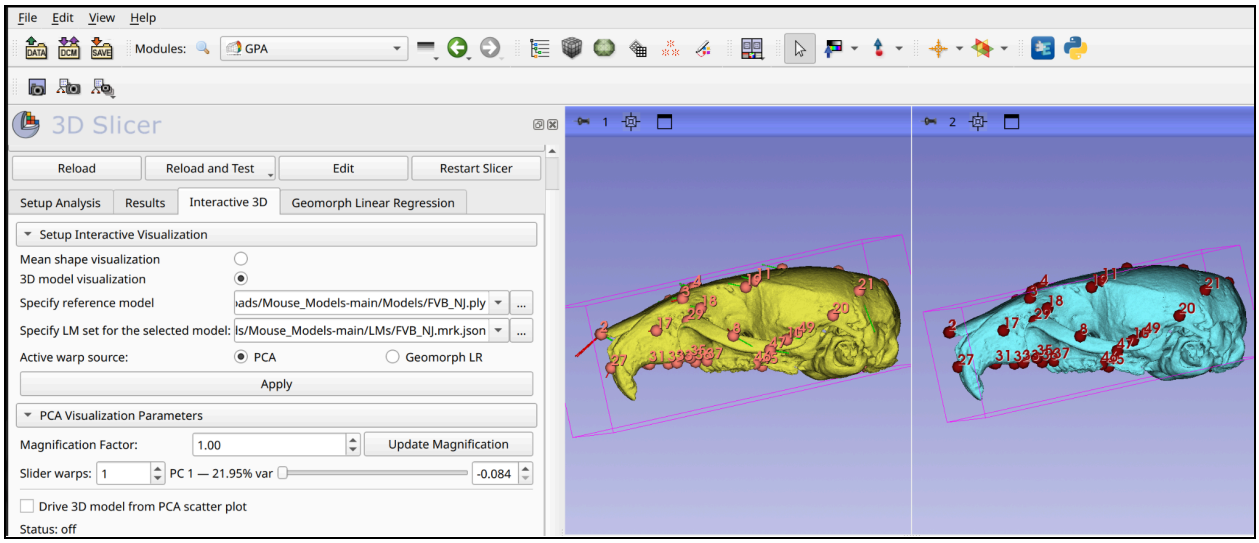
d. Look at landmark 14 (also discussed in 18.c.ii)

i. Inspect PC1 (red) and PC2 (green) vectors

1. Which direction does PC1 vary in?
    a. What morphological change could this correspond to?
2. Which direction does PC2 vary in?
    a. What morphological change could this correspond to?

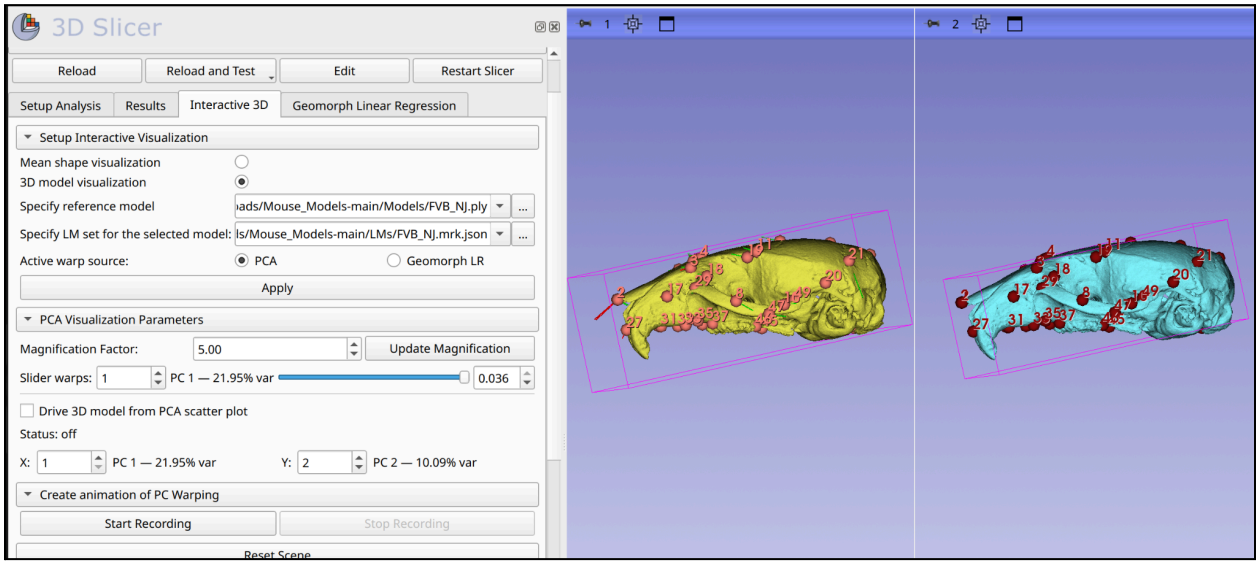
e. Click “Restore View Layout” icon to return to multipanel view

22. Return to the “Setup Analysis” tab
    a. Scroll down to “GPA Module Log Information”
    b. Check “Closest sample to mean” in log outputs
        i. We are going to load this specimen and landmarks into view in the “Interactive 3D” tab
            1. For this data, the specimen is FVB_NJ
23. Switch to the “Interactive 3D” tab
    a. Specify reference model: FVB_NJ.ply
        i. Or the path to your specimen closest to mean from **21.b** above
    b. Specify LM set for the selected model: FVB_NJ.mrk.json
        i. Or the path to your specimen closest to mean from **21.b** above
    c. Active warp source: PCA
    d. Click “Apply”

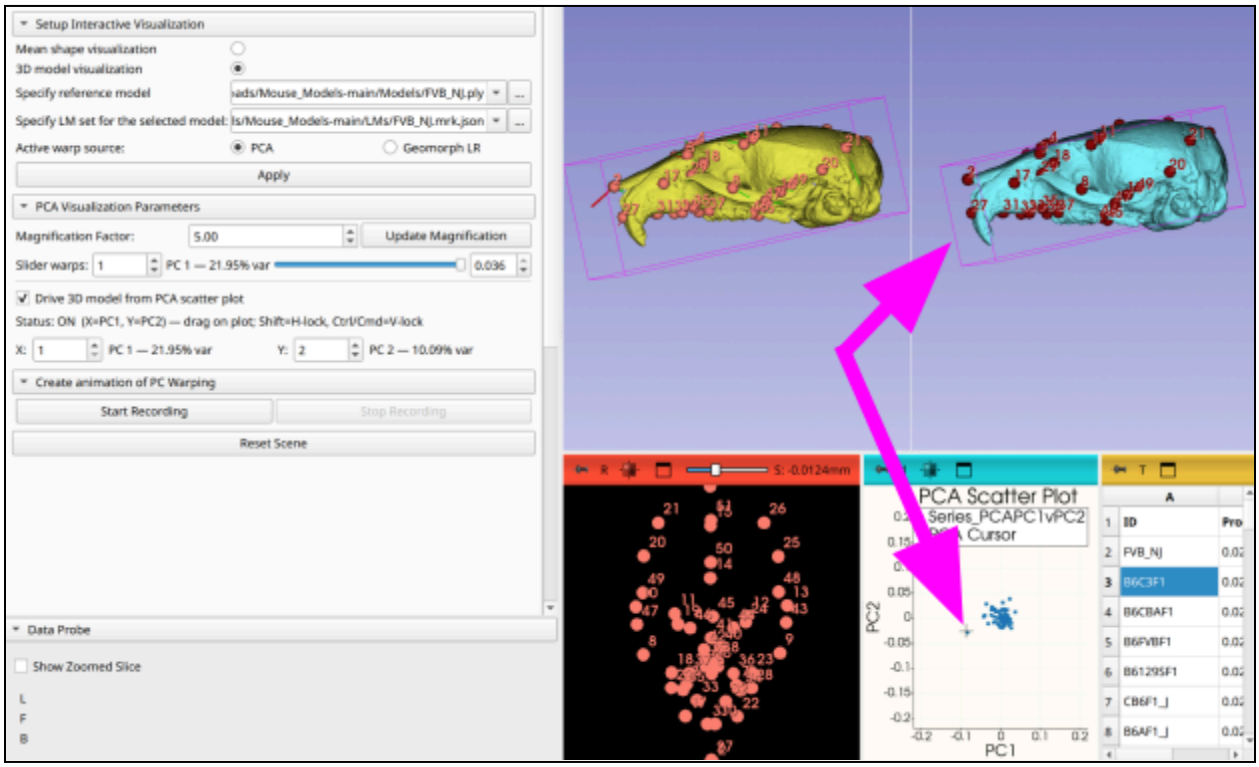


e. The left panel shows your reference model warped to the mean using the PCA mean landmark configuration
f. The right panel shows the actual reference specimen and landmarks

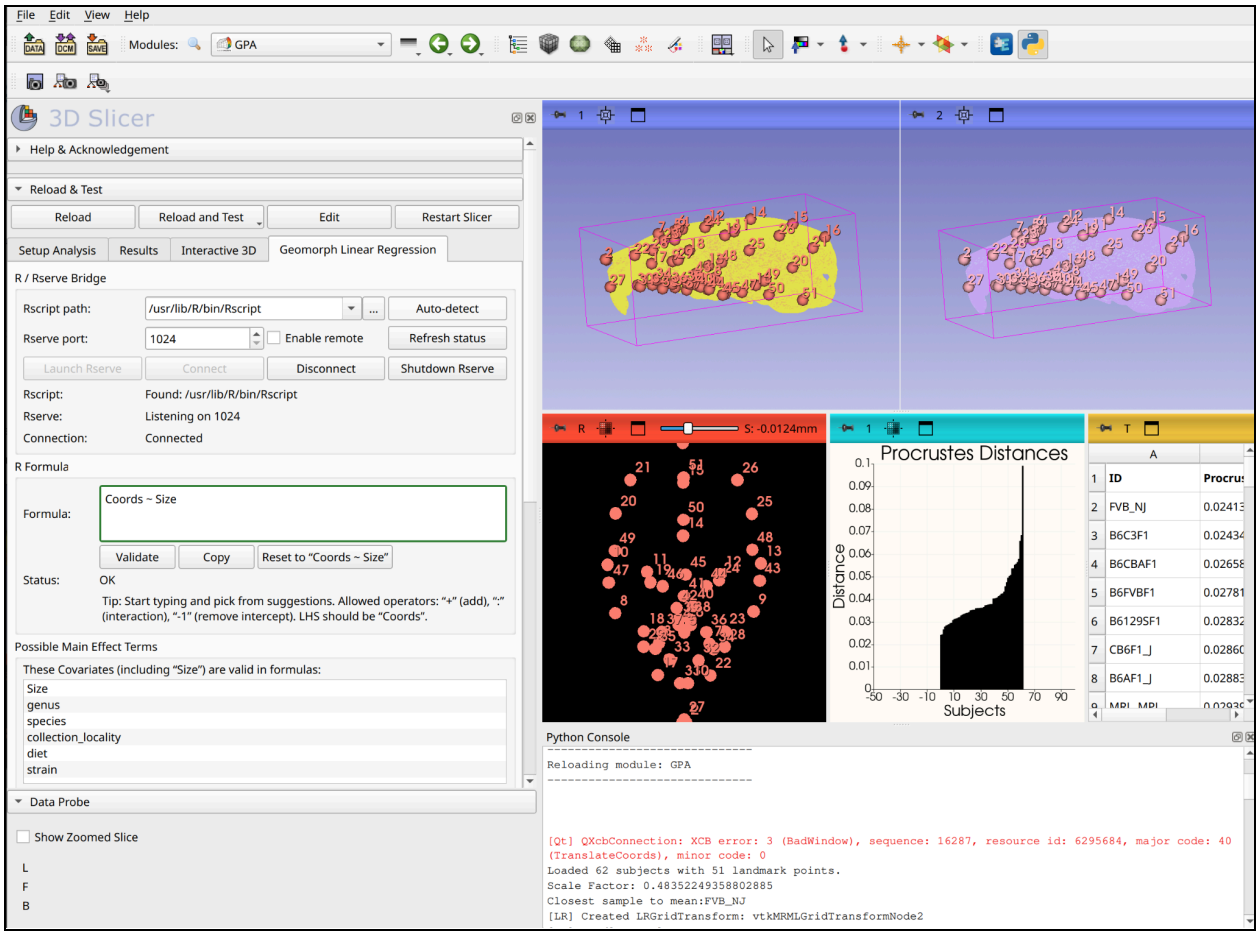
24. Adjust “PCA Visualization Parameters”
    a. **Watch the blue model warp along PCs using parameters you set here to understand morphospace trends**
    b. Magnification Factor: 10.00
        i. This updates the magnification of PCA warping to 3 SD’s from the mean
        ii. We are using a high value to make visualizing changes in lab mice easier. For most non-biomedical/model organism datasets, 1.00-3.00 is plenty
    c. Slider warps: 1
        i. This warps the specimen along PC1
    d. PC X Slider
        i. Adjust the slider to see min and max shapes along chosen PC
            1. **What shape changes do you notice across PC1?**



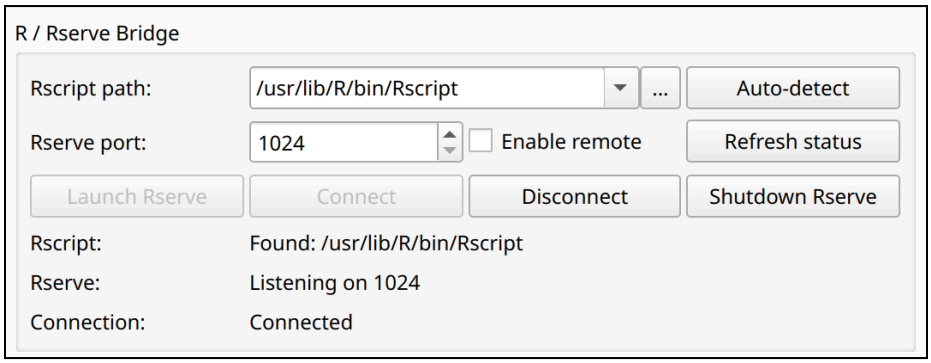
e. Optional
   i. Check “Drive 3D model from PCA scatter plot”
   ii. Click a point in the PCA Scatter Plot
   iii. Watch the blue (reference) model warp using the landmark configuration of the selected point



25. Under, “Setup Interactive Visualization”
    a. Active warp source: Geomorph LR
26. Switch to the “Geomorph Linear Regression” tab
    a. Rscript path: /usr/lib/R/bin/Rscript
        1. Point to your Rscript file
    ii. Click “Refresh status”
    iii. Click “Launch Rserve”
    iv. Click “Connect”

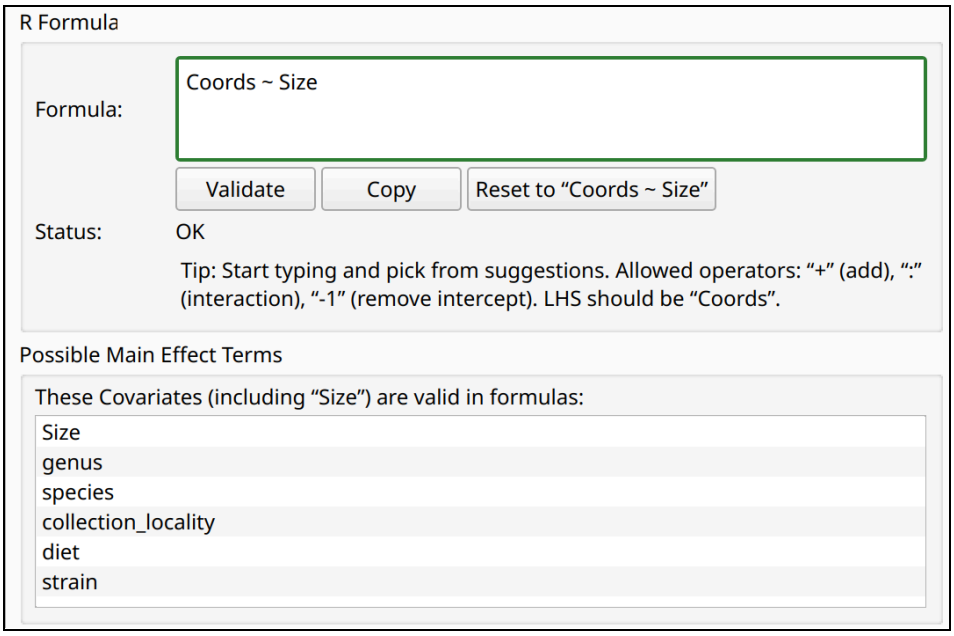


b. Check that status checks passed for Rscript, Rserve, and Connection

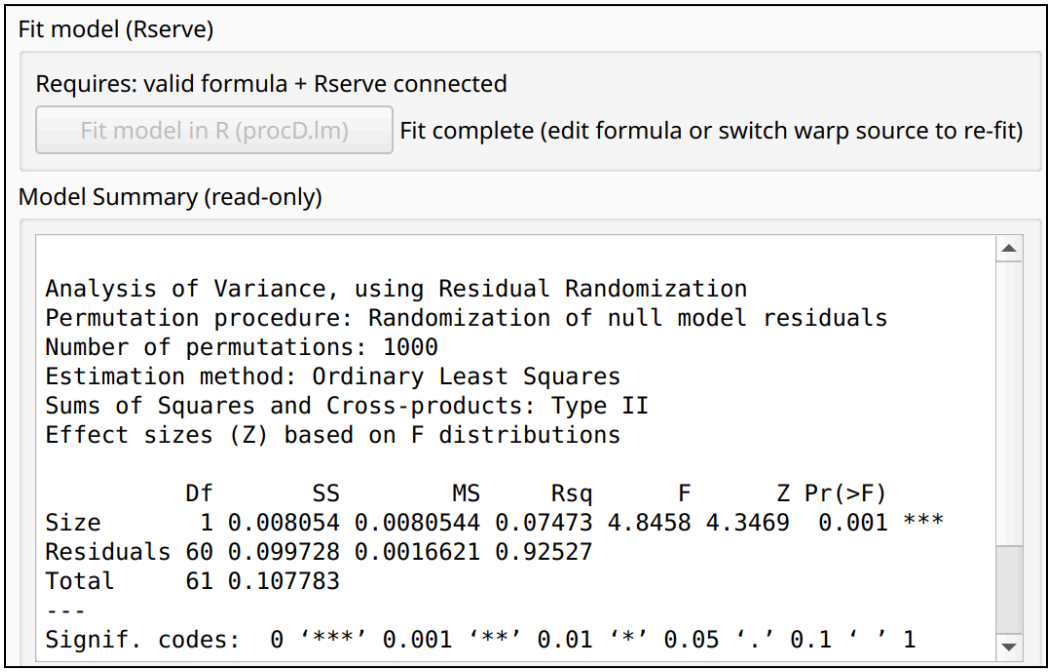


27. Start with a basic linear model in "R Formula"

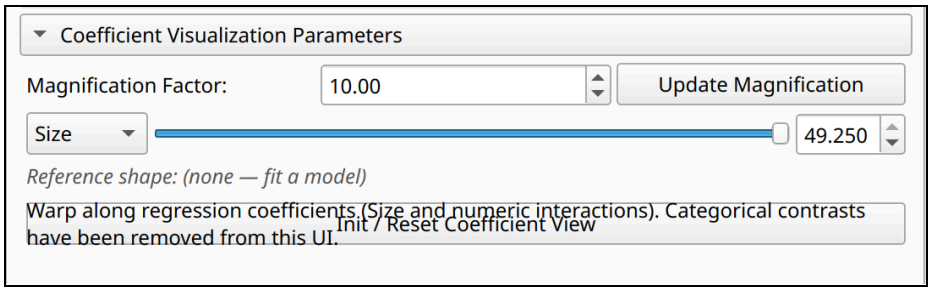
a. Coords ~ Size
   i. shape (the Procrustes coordinates, Coords) as a function of size
   ii. Null/base model: is shape explained by size alone?



b. Click "Fit model in R (procD.lm)



c. See the model as a shape change
    i. Go to "Coefficient Visualization Parameters"
        1. Magnification factor: 10.00
        2. Click "Update Magnification"
        3. Adjust the slider and view shape change for the warped mesh



28. Add a grouping variable to see if groups differ
    a. Coords ~ Size + collection_locality
        i. + fits main effects only
        ii. Common slopes: groups differ in mean shape but share the size trend
    b. Coords ~ Size * collection_locality
        i. * is shorthand for main effects plus their interaction
            1. Size * locality expands to Size + locality + Size:locality
        ii. Unique slopes: groups also differ in how shape changes with size
29. Congrats! You've completed the 3D GMM in 3D Slicer tutorial
    a. Be sure to check out SlicerMorph Tutorial for more information
        i. https://github.com/SlicerMorph/Tutorials/tree/main
30. To continue with some other classic geometric morphometric analysis from R geomorph (Adams et al. 2025), check out the demo on Google colab and the R script in the tutorial GitHub
    a. https://github.com/aubricot/Mouse_Model_GMM/